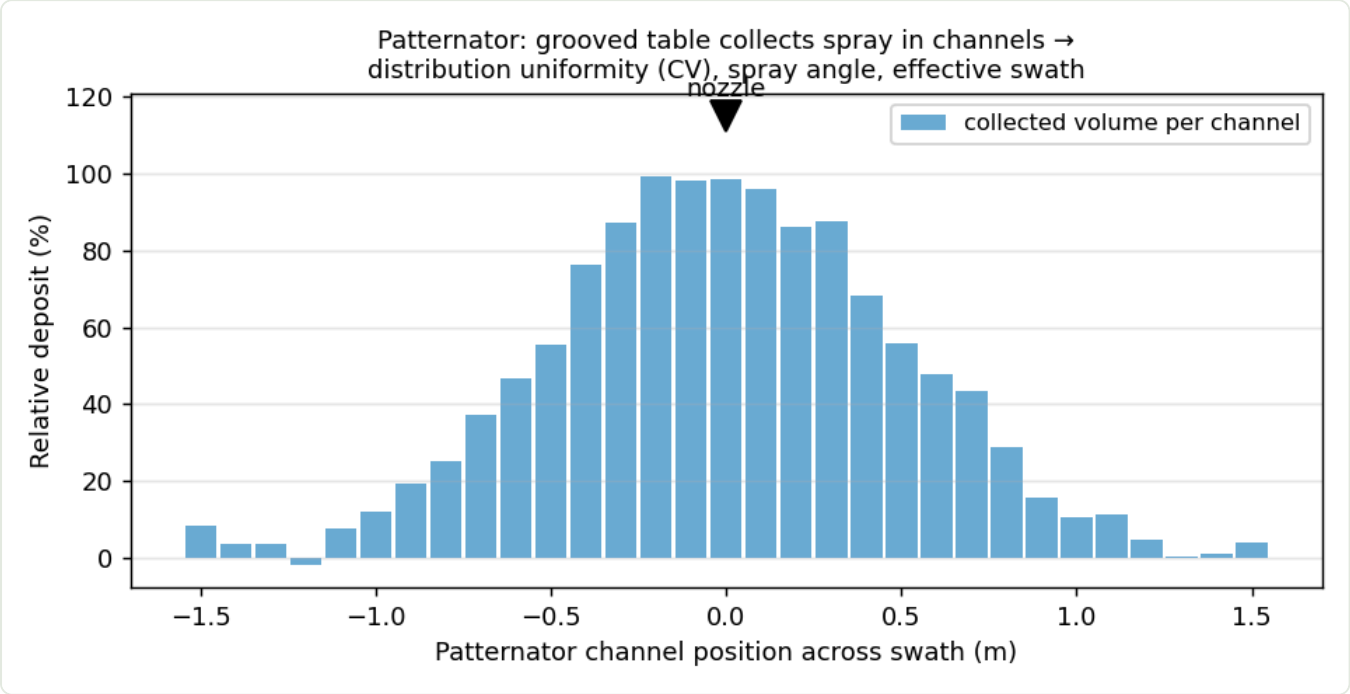
UNIT 3 · TOPIC 3(B)

Testing of Sprayers and Dusters

Syllabus topic: Testing of fertilizer applicators, sprayers and dusters

Laboratory tests

Component	Tests
Pump	Discharge vs pressure and rated speed; relief-valve function; power draw
Tank	Capacity; agitation (concentration uniformity kept within limits during run-out); strainers
Nozzles	Discharge rate (±5% of rated); SPRAY ANGLE (photograph/protractor); spray PATTERN on a PATTERNATOR (grooved table + graduated cylinders) — distribution across the swath, its CV, and the effective swath



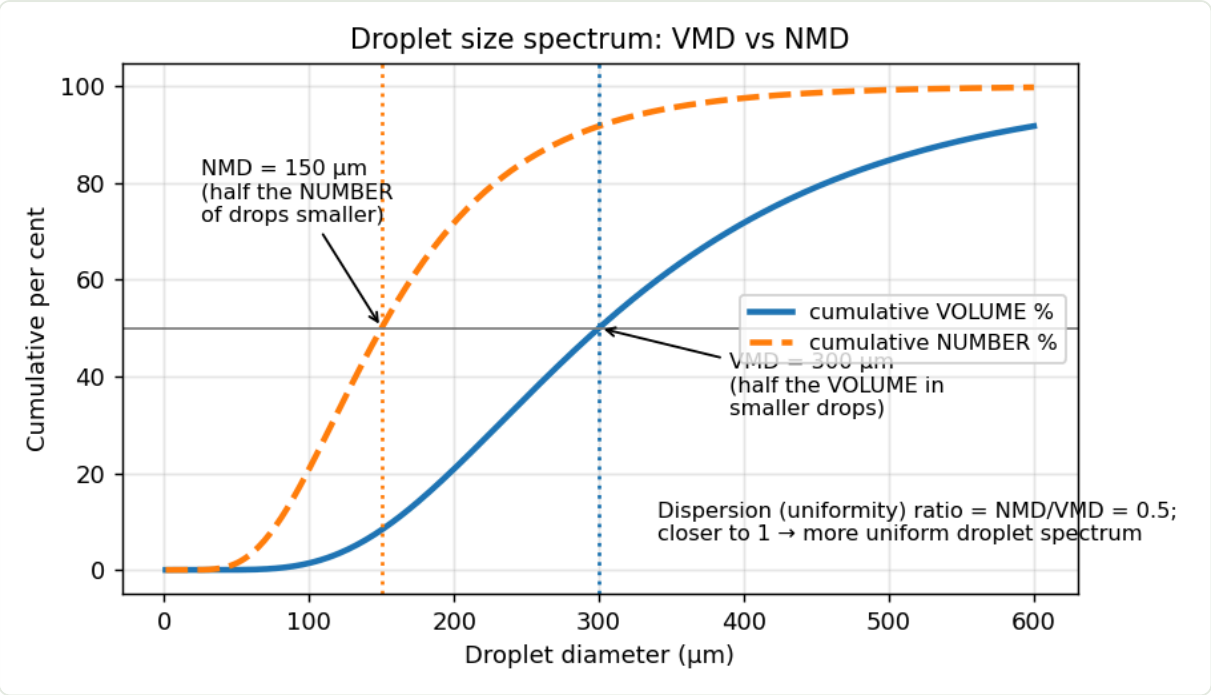
Patternator — distribution uniformity, spray angle and effective swath

Droplet spectrum

- Sample droplets on water-/oil-sensitive paper or glass slides (MgO coating) and size them by microscope/image analysis, or measure them by laser diffraction.
- **VMD** (volume median diameter, D_v0.5): half the sprayed VOLUME is in drops smaller than the VMD.
- **NMD** (number median diameter): half the NUMBER of drops are smaller.
- A few big drops carry most of the volume, so **VMD > NMD always**.

$$\text{Dispersion (uniformity) ratio} = \text{NMD/VMD} \text{ — closer to } 1 = \text{more homogeneous spectrum}$$

Two conventions are in use: NMD/VMD (≤ 1, used here) and VMD/NMD (≥ 1, used in Unit 1). Both indicate a uniform spectrum as they approach 1 — read the MCQ options carefully.



Droplet size spectrum — VMD vs NMD

Target	Droplet density (indicative)
Herbicides	20–30 drops/cm²
Insecticides	20–30 (systemic) to 30–40 drops/cm²
Fungicides	50–70 drops/cm² (protectants need dense cover)

Droplet class	Size and use
Fine	100–200 µm — drift-prone
Medium	200–300 µm — most field spraying
Coarse	>300 µm — herbicides, drift control

Field tests

- Verify the application rate over a measured area.
- Measure swath and uniformity with tracer dye/fluorescent tracer and paper targets in the canopy (deposit at top/middle/bottom).
- Assess drift downwind; operator exposure/safety (Unit 5); tank run-out concentration stability.

Dusters

- Feed rate (g/min) vs metering opening — by weighing.
- Distribution across the swath with trays; uniformity vs hopper level.
- Blower air velocity — anemometer.

Practice MCQs

Q58. Spray distribution across a nozzle's swath is measured on a:

(a) Patternator (b) Profilometer
(c) Anemometer (d) Planimeter

Q59. Nozzle discharge in a test should be within:

(a) ±1% (b) ±5%
(c) ±15% (d) ±25% of rated

Q60. VMD is the diameter such that:

(a) Half the drops are smaller (b) Half the spray VOLUME is in smaller drops
(c) All drops are equal (d) Drift is zero

Q61. NMD relative to VMD is always:

(a) Larger (b) Equal
(c) Smaller (d) Random

Q62. Uniformity (dispersion) ratio of a spray is:

(a) VMD/NMD (b) NMD/VMD
(c) VMD×NMD (d) VMD−NMD

Q63. A dispersion ratio approaching 1 indicates:

(a) Very non-uniform drops (b) Homogeneous droplet spectrum
(c) Coarse spray only (d) High drift

Q64. VMD 300 µm and NMD 150 µm give a uniformity ratio of:

(a) 0.5 (b) 1.0
(c) 2.0 (d) 4.5

Q65. Droplets are commonly sampled on:

(a) Litmus paper (b) Water-/oil-sensitive paper or coated slides
(c) Filter paper only (d) Cloth

Q66. Recommended droplet density for protectant FUNGICIDES is about:

(a) 5–10 (b) 20–30
(c) 50–70 (d) 200–300 drops/cm²

Q67. 'Fine' spray droplets (100–200 µm) chiefly increase:

(a) Runoff (b) Drift risk
(c) Draft (d) Corrosion

Q68. Canopy deposit distribution in field sprayer tests is assessed using:

(a) Tracer dye/fluorescent tracer on targets (b) Odometer
(c) Thermometer (d) pH strips

Q69. Duster feed rate is calibrated by:

(a) Volume in jar (b) Weighing output per unit time
(c) Counting particles (d) Pressure gauge

Answers

Q58 — (a) Patternator

Q59 — (b) ±5%

Q60 — (b) Half the spray VOLUME is in smaller drops

Q61 — (c) Smaller — *A few large drops hold most of the volume, so the number median lies well below the volume median.*

Q62 — (b) NMD/VMD
Q63 — (b) Homogeneous droplet spectrum
Q64 — (a) 0.5 — *Ratio = 150/300 = 0.5 — a typical hydraulic-nozzle value.*
Q65 — (b) Water-/oil-sensitive paper or coated slides
Q66 — (c) 50–70
Q67 — (b) Drift risk
Q68 — (a) Tracer dye/fluorescent tracer on targets
Q69 — (b) Weighing output per unit time

Sources: RNAM (UN-ESCAP) Test Codes and Procedures for Farm Machinery; BIS Indian Standards for seed drills, sprayers and dusters; Kepner et al. (performance concepts); Jagdishwar Sahay; FMTTI (Budni) test procedures; NPTEL lectures.
